A Candidate Degree-494 Skein-Lasagna Obstruction for the Cappell–Shaneson Sphere X(41,189,73)

Abstract. We present a candidate obstruction to the smooth four-dimensional Poincaré conjecture for the Cappell–Shaneson manifold X(41,189,73). The proposed distinguishing class lies in homological degree zero and quantum degree 494 of the N=2 skein-lasagna module over Q. A divided cubic functional evaluates on the selected class as −59072, while the standard four-sphere module vanishes in that quantum degree. The argument constructs a balanced one-handle class, descends its detector through the full beta/psi relations and six relations from a chosen three-sphere basis, and then invokes the published four-handle and standard-sphere comparisons. Exact finite calculations, immutable geometry certificates, and a Lean kernel audit are supplied. The Lean development checks the arithmetic and the abstract quotient implication; candidate-specific geometric identifications remain explicit interfaces and therefore require independent human verification. This draft is not peer reviewed.

Keywords. Smooth four-dimensional Poincaré conjecture; Cappell–Shaneson sphere; skein-lasagna module; Khovanov–Rozansky homology; handle decomposition; formal verification.

1. Introduction

The smooth four-dimensional Poincaré conjecture asks whether every smooth, closed four-manifold homotopy equivalent to S^4 is diffeomorphic to S^4. The Cappell–Shaneson construction produces a large and unusually explicit family of homotopy four-spheres from matrices in SL(3,Z). Many members of the family have been shown standard by Kirby-calculus arguments, but the presentation remains a natural testing ground for invariants that can distinguish smooth structures.

This paper studies the member X(41,189,73). The proposed obstruction is not a conventional numerical invariant attached directly to the matrix. It is a nonzero class in a graded skein-lasagna module, detected only after a coefficient trace, a point-push operator, and the complete handle relations have been accounted for. The calculation is delicate because a nonzero number before the full quotient has no distinguishing force: the beta, psi, and three-handle relations may kill the class.

> **Main claim (candidate proof).** *Subject to the published handle-decomposition theorems cited in Section 3 and the candidate-specific geometric identifications established in Sections 4–7, X(41,189,73) is a homotopy four-sphere and is not diffeomorphic to the standard S^4. Consequently it would be a counterexample to the smooth four-dimensional Poincaré conjecture.*

The qualifier 'candidate proof' is substantive. The complete natural-language argument and all finite certificates are public, but the Lean theorem is conditional on explicit geometry records. A clean axiom report does not discharge those records, because theorem parameters are not Lean axioms. Section 9 states this boundary precisely.

1.1. Structure of the argument

- Construct a balanced one-handle class v_T and a divided cubic functional D_3.
- Compute $D_3(v_T)=-59072$ and show that v_T has final quantum degree 494.
- Prove that D_3 kills every beta/psi relation and the six relations supplied by three chosen embedded spheres.
- Use the complete MWW quotient and four-handle isomorphism to obtain a nonzero degree-494 class of X(41,189,73).
- Use the standard-sphere calculation, which is concentrated in quantum degree zero, to exclude a diffeomorphism with S^4.
- Use the Cappell–Shaneson determinant criterion and standard topological arguments to identify the candidate as a homotopy sphere.

2. The Cappell–Shaneson candidate

The matrix used throughout is

$$A = [[0, 269, 1240], [0, 41, 189], [1, 0, 32]]. \quad (2.1)$$

Exact integer arithmetic gives det(A)=1 and det(A−I)=1. The first identity places A in SL(3,Z); the second is the standard Cappell–Shaneson condition ensuring that the associated mapping-torus surgery has the homology and fundamental-group behavior required of a homotopy sphere. The Lean files T73Finite.lean and T73CSAlgebra.lean verify the matrix arithmetic and explicit inverses for the maps A−I and Λ^2 A−I.

The topological promotion is logically separate from the smooth obstruction. Van Kampen supplies simple connectivity from the surjectivity of A−I. The Wang and Mayer-Vietoris sequences, together with Poincaré duality, give the integral homology profile of S^4. Hurewicz and Whitehead then promote the result to a homotopy equivalence. In the Lean development these implications are represented by three corresponding fields of CSTopologyData; the candidate-specific applications are not presently proved inside the kernel.

3. Skein-lasagna modules and external results

Let S^2_0(X;Q) denote the N=2 skein-lasagna module over Q, with its homological and quantum gradings. The proof uses the handle presentation of Manolescu, Walker, and Wedrich (MWW): a direct sum of raw cabled state spaces is quotiented first by beta/psi relations and then by the relations associated to three-handle attaching spheres. A four-handle induces a grading-preserving isomorphism.

Input	**Role in this paper**	**Status**
MWW complete handle quotient	Identifies the full beta/psi and sphere relation space; supplies the universal quotient.	Published theorem; applicability must be checked.
Horvat–Jabłonowski replacement	Permits a pairwise-disjoint unimodular sphere basis to replace the historical three-handle system.	Published/preprint theorem; candidate hypotheses checked by finite geometry.
MWW four-handle map	Carries the surviving degree-494 class to the closed four-manifold.	Published theorem.
MWW S^4 computation	$S^2_0(S^4;Q)$ is concentrated in bidegree (0,0).	Published computation; rational scope retained explicitly.
BPW/BHPW trace functoriality	Carries the coefficient Hattori class to the endpoint shadow and makes the required ordinary cobordism maps strict.	Published categorical input.
Cappell–Shaneson topology	Turns the determinant conditions into the homotopy-sphere conclusion.	Classical topology plus the cited modern restatement.

No appeal is made to a conjectural functoriality for knotted webs or singular foams embedded in four-space. The geometric consumers in the argument are ordinary framed oriented surface cobordisms between tangles or links, together with the published horizontal- and vertical-trace constructions.

4. The balanced class and the divided cubic detector

4.1. Raw state and Hattori class

For a finite cabled state s=(α,r), let C_s be the corresponding raw MWW summand. At the selected state s_0, the active coefficient boundary has one negative and one positive cable on each of m_2 and r_xy. Thus α=0, r is the sum of the two corresponding basis vectors, and |r|=2.

$$C(s) = S^2_0(W_1; K(r-\alpha^-, r+\alpha^+) \cup L, \eta^r; Q)\{-(2|r|+|\alpha|)\}. \quad (4.1)$$

The actual paired-annulus cut yields an action-compatible coefficient equivalence. Here B_act is the framed automorphism of the 88-endpoint object induced by the paired-annulus motion. The two transported boundary objects are T_1=B_act^{-1}U and T_0=B_act^{-1}WU, where U is the physical cup and W is the point-push braid. The selected diagonal Hattori class is

$$v_T = \eta_R[T_1] = [\ H^{-1}(\mathrm{Id}_U \otimes X^{\otimes 227})\ ]. \quad (4.2)$$

The 227 symbols X label separate Frobenius-algebra factors; they are not the vanishing product X^{227}. The coefficient trace and 227 counits carry this class to the endpoint vector u=e_0−e_5. The discarded alternative $\xi=\eta_R[T_0] - s_{inv}\eta_R[T_1]$ is not used: applying the same outer point-push difference to ξ introduces a second factor and first contributes in order six.

4.2. Quantum trace and divided functional

Let ζ be the quantum-trace parameter and complete at $\zeta=1+h$. The point-push word lies in the third term of the relevant filtration, so $\rho\_h(W)-I$ is divisible by h^3 on the full endpoint module. With a normalized cap $\hat{C}(h)$, division by h^3 followed by reduction modulo h defines an ordinary linear functional D_3 on coefficient HH_0. The map Θ_h is the completed quantum-Hochschild shadow from the actual coefficient module to the endpoint module.

$$D_3 = [h^3]\ \hat{C}(h)\ (\rho\_h(W)-I)\ P_{86}\ \Theta\_h. \quad (4.3)$$

The projector P_{86} isolates the one-cup top cell. It is applied only after the genuine quantum-Hochschild shadow; it is not asserted to be a physical foam. The construction is lift-independent because changing a lift by h y changes the divided value only by a multiple of h.

4.3. Exact point-push calculation

Put $t=\zeta^{-2}$ and $\varepsilon=t-1$. Exact integer arithmetic gives

$$\ell(\rho(W)-I)u = 7384\ \varepsilon^3 - 660412\ \varepsilon^4 + 34814626\ \varepsilon^5 - 1365512573\ \varepsilon^6. \quad (4.4)$$

The coefficients in degrees zero, one, and two vanish. Since $\varepsilon=-2h+3h^2-4h^3+O(h^4)$, the cubic coefficient is

$$D_3(v_T) = (-2)^3 \cdot 7384 = -59072 \neq 0. \quad (4.5)$$

The public recomputation starts from 252 primitive crossing rows and a chronology record. It reconstructs an 11,340-letter B44 word, then its 45,360-letter two-cable word, and performs the truncated Burau action. Neither 7384 nor −59072 appears in the input JSON. The value belongs to the registered oriented point-push chronology. A different, non-chronological ordering of the same crossing rows gives −58976 and moves the first anomaly from degree three to degree two. The legality and naturality arguments must therefore establish invariance under every permitted change of presentation; the finite number alone is not yet a manifold invariant.

4.4. Grading

The four grading contributions are exact:

$$-44 + 227 + 315 - 4 = 494. \quad (4.6)$$

Hence the proposed surviving class lies in bidegree (0,494).

5. Descent through the one- and two-handle relations

5.1. One-cup firewall

The endpoint Bar-Natan/Temperley-Lieb category is filtered by through degree: F_k is the span of diagrams with through degree at most k, and composition cannot raise this filtration. The physical cup U has through degree 86. Every undotted balanced-pair creation for an active gate-crossing owner contains at least two cups, so its full action-closed ideal lies in F_{84}. The rational projector P_{86} therefore kills that ideal while preserving the selected vector.

The zero-gate owner r_zx is handled separately. After the artificial meridian spectator is deleted, its balanced pair is a split zero-framed unknot. For the standard rank-two Frobenius algebra $A=Q[X]/(X^2)$,

$$(\varepsilon \otimes \varepsilon)\ \psi^{(0)} = 0,\ (\varepsilon \otimes \varepsilon)\ \psi^{(1)} = \mathrm{Id}. \quad (5.1)$$

5.2. All finite cable states

At the base state the constant balanced-pair quotient is trivial and every pure generator acts as I+O(h). At higher multiplicities, physical copies are put into a fixed order by stable positive shuffles, namely positive braids that preserve the order already assigned to existing copies. After the actual W2 core disks are attached, the functional is averaged over the finite permutation orbit; this is the Reynolds average used here. The normalization ratios for successive orbit sizes telescope, so the intermediate factors cancel. Distinct-owner squares commute because their supports are disjoint. Consequently, for every finite state and every beta/psi edge,

$$\Lambda_t^{(3)}\ \Psi_e^{\ 0} = 0,\ \Lambda_t^{(3)}\ \Psi_e^{\ 1} = \Lambda_s^{(3)}. \quad (5.2)$$

Equations (5.1)-(5.2) imply that the divided functional annihilates the complete one- and two-handle relation subspace. This is an all-state statement at order h^3, not a test on the selected vector alone.

6. A chosen three-sphere basis

6.1. Finite geometry

The historical attaching-sphere movies are not recovered and are not used. Instead, the proof constructs three new chosen spheres, denoted TH1, TH2, and THXY, in the same post-two-handle boundary. Their frozen certificates and hostile re-audits are included in the public evidence directory.

Sphere	Construction counts	Frozen SHA-256 prefix
TH1	350,176 leaves; 350,175 bands; one root cap	EE620E6B085A5F9...
TH2	229,198 leaves; 229,197 bands; one root cap	4D1B627C0343A1C...
THXY	11,115 material disks; 11,114 split bands; one root cap	EABF67C0D1AE309...

The certificates record complete ancestry hashes, embedded product-normal bands, root-cap consumption, and Euler-characteristic ledgers ($\chi = 2$, i.e. genus zero). The three movie-time (height) intervals are disjoint: THXY lies in [2,13/2], TH1 in [8,9], and TH2 in [10,11], while the old one-cup/core block lies in [−4,−3].

6.2. Homology basis and replacement

Their spherical homology coordinates are

$$v_1=(-1311,8608,-1),\ v_2=(-189,1241,0),\ v_3=(41,-269,1). \quad (6.1)$$

$$\det[v_1\ v_2\ v_3] = 1. \quad (6.2)$$

The Horvat–Jabłonowski basis theorem is then invoked to replace the historical three-handle attaching system by this pairwise-disjoint unimodular basis, up to permutation and three-handle slides. The external link L is empty in the present application. This replacement theorem is an external published/preprint input; the embeddedness, disjointness, and determinant-one hypotheses are candidate-specific finite claims.

6.3. The six sphere relations

For each chosen sphere j, every noninvertible critical point lies in a new material factor. The old one-cup block is transported by identity cylinders. After the fixed actual-to-canonical coordinate maps, the constant endpoint maps have the form

$$F(j,0)^{(0)} = Q(t\_j,0)^{-1}\ (\mathrm{Id\_old} \otimes U_j)\ Q(s,0), \quad (6.3)$$

$$F(j,1)^{(0)} = Q(t\_j,0)^{-1}\ (\mathrm{Id\_old} \otimes D_j)\ Q(s,0), \quad (6.4)$$

$$D_j=X^{\otimes b_j};\ U_j=\sum_{a=0}^{b_j-1} X^{\otimes a} \otimes 1 \otimes X^{\otimes(b_j-1-a)}. \quad (6.5)$$

The W2 core disks give E_j as the b_j-fold tensor power of the counit ε, whence $E_j(U_j)=0$ and $E_j(D_j)=1$. Write $\Psi^{quot}(j,n)$ for the map induced after the beta/psi quotient. Therefore

$$\ell_1\ \Psi^{quot}(j,0)=0,\ \ell_1(\Psi^{quot}(j,1)-\mathrm{Id})=0,\ j=1,2,3. \quad (6.6)$$

These are identities of linear maps on the whole source under the stated naturality square. The stored scalar pairs 0/0, 0/0, 0/0 are consequences, not substitutes for (6.6). Positive transport corrections start at h^4 after multiplication by the old h^3 anomaly and therefore do not alter the divided cubic. This order estimate concerns endpoint transport. It does not by itself settle a change in the insertion position of a collar word; that separate geometric binding is listed in Section 11.

7. Passage to the closed manifold

We use three levels of notation. D_3 denotes the numerical value of the detector on the selected class; Λ_3 denotes the compatible functional on the full direct sum of raw states; Λ_3^{quot} denotes its descent to the complete quotient. The Lean fields ℓ_0, ℓ_1, and ℓ_2 are the corresponding functionals at the successive quotient stages.

Let C be the direct sum of all finite raw cable states, R_2h the complete beta/psi relation subspace, and R_3h the six-relation subspace from the chosen sphere basis. Sections 5 and 6 assert

$$R_2h \subset \ker \Lambda_3,\ R_3h \subset \ker \Lambda_3. \quad (7.1)$$

Write $v\_raw(s_0)$ for the raw-state representative of v_T in the selected state s_0.

The quotient universal property produces

$$\Lambda_3^{quot} : C/(R_2h+R_3h) \to Q,\ \Lambda_3=\Lambda_3^{quot}\ \pi. \quad (7.2)$$
$$\Lambda_3^{quot}(\pi(v\_raw(s_0))) = -59072 \neq 0. \quad (7.3)$$

Thus the selected degree-494 class survives the complete handle quotient. The published four-handle isomorphism preserves its bidegree and nonvanishing. For the standard four-sphere, the same invariant is concentrated in bidegree (0,0); in particular $S^2_{0,494}(S^4;Q)=0$. Graded diffeomorphism invariance would carry a degree-494 class of the candidate to one of S^4, which is impossible. The standard control itself has W_std=I and thus checks only a zero-to-zero map; it cannot independently diagnose a failure of the candidate descent.

Smooth obstruction. *If the actual-geometry identifications, complete MWW quotient, chosen-sphere replacement, four-handle comparison, and rational S^4 control have the scopes stated above, then X(41,189,73) is not diffeomorphic to S^4.*

8. Evidence and independent replay

The public repository separates proof prose, exact finite inputs, replay programs, and Lean sources. The initial public release mistakenly retained several load-bearing geometry objects only by local path and SHA-256. Commit 9797850472602f311b6957244048044d70b1adb4 corrects that omission.

Layer	Public artifact	What can be independently checked
Point-push cubic	T73_DELTA3_PUBLIC_INPUT.json; recompute_t73_delta3.py	Rebuild B44/B88; exact ε series; −59072.
G1 global descending	t73_reduced_billiard.pd.json; verify_t73_global_descending.py	All 2,126,291 crossings, components, basepoints, height order.
E1 coefficient cut	ACTUAL_PD_CABLE_UNIT_CERT.json	Exact passage and disk certificate; frozen input identity.
E8 chosen spheres	TH1/TH2/THXY JSON plus hostile re-audits	Actual frozen certificate contents, hashes, counts, determinant and scalar fields.
Lean kernel	Smooth4PC/*.lean; T73Audit.lean	Arithmetic, abstract trace/quotient algebra, final conditional implication, axiom reports.

The public geometry verifier checks 29 immutable files and recomputes the global-descending certificate from the full PD. It does not regenerate the three chosen-sphere certificates from their much larger upstream construction trees. Those JSON files are now available for independent inspection, but their geometric semantics remain part of the human proof rather than the Lean kernel.

9. What the Lean development proves

9.1. Actual mathlib use

The project pins mathlib at revision 520045ab14e26149ee970e2e617ca04b09bde5d6. It imports rational linear algebra, quotient modules, finite sums, the fundamental-groupoid simply-connectedness predicate, and singular homology. The project does use mathlib; the substantive question is whether the candidate-specific geometry has been constructed as Lean data.

9.2. Kernel-checked content

- The determinant and lattice calculations for A, A−I, and Λ^2 A−I.

- The exact arithmetic identities computedCubic=−59072 and computedDegree=494 from frozen integer inputs.
- Generic coefficient-HH_0 quotient and descended-row naturality lemmas.
- The positive split-tree Frobenius identities: the undotted row is zero and the dotted row is the source row.
- Generic sphere-relation kernel inclusion and the implication from a nonzero descended row to a nonzero quotient class.
- The final logical implication from an ExternalGeometry and Cappell–Shaneson topology package to 'homotopy sphere and not diffeomorphic to S^4'.

9.3. Explicit formalization boundary

The final theorem has the schematic type

```
conditionalCounterexample
(geom : ExternalGeometry u)
(cs : CSExternalGeometry u) :
IsHomotopySphere candidate /\ not Diffeomorphic candidate S4
```

No tracked file constructs geom or cs. ExternalGeometry contains W0-W3, the selected class and functional, the equation $\ell_0(x0)$=−59072, quotient maps q_{01} and q_{12} with compatibility equations, a transport equivalence, the four-handle equivalence, vanishing of the standard-sphere degree, and graded diffeomorphism invariance. Related interfaces require actual sphere embeddings, class coordinates, six actual sphere maps, pairwise disjointness, HJ replacement, the MWW coequalizer, and the four-handle binding.

Likewise, CSTopologyData uses genuine mathlib notions of simply connectedness and singular homology but takes the candidate-specific van Kampen, Wang/Mayer-Vietoris/Poincaré, and Hurewicz/Whitehead implications as fields. The local file named Unconditional.lean contains documentation only and expressly says that the current result remains conditional.

9.4. Why the axiom audit is necessary but insufficient

A clean '#print axioms' report establishes that the proofs do not use sorryAx or undeclared project axioms. The fresh replay produces 38 reports, all contained in {propext, Classical.choice, Quot.sound}. This is useful kernel evidence. It does not show that ExternalGeometry is inhabited: an explicit theorem parameter is an assumption in the theorem type, not an axiom reported by '#print axioms'.

10. Reproduction protocol

At repository commit 9797850472602f311b6957244048044d70b1adb4, a reader may run:

```
git clone https://github.com/toffee-desuwa/smooth4pc-t73-lean.git
cd smooth4pc-t73-lean
lake update
git diff --exit-code -- lake-manifest.json
lake exe cache get
python -B tests/test_t73_minimal_formalization.py -v
lake lean T73Audit.lean
python -I -B scripts/recompute_t73_delta3.py --check
python -I -B scripts/verify_public_geometry_evidence.py
```

The expected outputs are: two Python formalization tests pass; T73Audit exits zero; 38 axiom reports contain only the three foundational axioms listed above; the point-push replay prints DELTA3_ETA_T1=−59072; and the geometry replay prints PD_CROSSINGS=2126291, GLOBAL_DESCENDING=PASS, and VERIFY=PASS.

11. Independent-review priorities

The finite arithmetic is the least uncertain part of the argument. A serious review should concentrate on the interfaces that connect finite certificates and published theorems to the actual smooth four-manifold. The following questions are decisive.

- Does the actual paired-annulus cut give the claimed two-sided Hattori coefficient, including framing and all left/right actions?

- Does the quantum vertical-to-horizontal trace apply with the stated grading and strict naturality to every ordinary cobordism used here?
- Does the actual-Gompf-to-DIAGRAM bridge preserve the complete labelled framed link, rather than only relator words or unframed projections?
- Do the TH1, TH2, and THXY certificates define three embedded, framed, pairwise-disjoint spheres in one actual W2, with the displayed homology coordinates?
- Does the Horvat–Jabłonowski replacement theorem apply with exactly these hypotheses and preserve the upper cobordism relative to W2?
- Are the six actual sphere maps identified with the canonical new-factor split maps on the whole source, not merely on the selected vector?
- Does the complete MWW quotient contain no additional relation that can kill the class?
- Are the four-handle and rational S^4 comparisons grading-preserving in precisely the form consumed by the proof?
- The value $D_3=-59072$ is computed for the registered oriented point-push presentation and serialization. Reordering the same crossing data can give −58976 and move the first anomaly from degree three to degree two. Which equivalence theorem proves invariance under every legal change of presentation and ordering?
- The standard S^4 control has W_std=I and hence sends zero to zero. Because that control cannot expose a failure of descent, is the candidate descent proved independently rather than inferred from agreement with the control?
- Is the insertion position of each sphere movie relative to the detector fixed by the geometric binding? A collar ambiguity $w \to w\,p$ with p in the third term Γ_3 of the lower central series of the pure braid group can be visible at order h^3; the observed value is divisible by 16, so the insertion-point issue cannot be dismissed by the present congruence alone.

A failure at any of these points breaks the claimed smooth obstruction even though all current Lean files continue to compile. Conversely, verifying these identifications independently would convert the present conditional kernel into strong support for the full mathematical argument.

12. Conclusion

The proposed counterexample is detected by a sharply localized discrepancy: a degree-494 class whose divided cubic value is −59072. The calculation is exact, the complete finite evidence is now public, and the abstract descent implication has been checked in Lean. What remains outside the kernel is not a vague appeal to intuition but a finite list of named geometric and published-theorem interfaces. This separation is the principal value of the present package: it makes clear both why the candidate would falsify the conjecture and exactly where an independent mathematical review must succeed or find an error.

Appendix A. Formal interface inventory

Interface field	Mathematical content	Current support
ℓ_0_x0	The selected actual class evaluates to −59072.	Exact Python replay plus human binding; explicit Lean field.
q_{01}, q_{12}	Complete two successive quotient maps.	Abstract quotient lemmas in Lean; actual identification external.
ℓ_1_comp_q_{01}, ℓ_2_comp_q_{12}	Detector descends through both relation spaces.	Generic Lean implication; candidate naturality human-checked.
transport, fourIso	MWW transport and four-handle comparison.	Published input; explicit Lean field.
s4DegreeZero	The S^4 group in degree 494 vanishes.	Published computation; explicit Lean field.
diffeomorphismEquiv	Diffeomorphism induces a graded equivalence.	Invariant functoriality; explicit Lean field.
sphere bindings	Actual embedded spheres induce the six claimed maps.	Public JSON/prose evidence; not instantiated in Lean.
CSTopologyData	Simple connectivity, homology-sphere profile, and homotopy promotion.	Mathlib definitions; candidate implications supplied as fields.

Appendix B. Core immutable identities

```
T73 delta input 04507506DC577384BC4C04765CCB212C1481DC71810C6AB3232F1AB690F16909
B44 derived word 7C2D2F792C2672221A76CAF08A71F560AF0CB7654B4D537C00FAD00B16EFA187
B88 derived word C13D0A3BA9B05F41A6D2C5B4AB12DDECDABEFC1883835891AD3BD2B8955B5FFB
Actual cable unit 7F3D3618D6A790A9B60EE8085B647AC2AB742E1BC9C15841F1BEF015034217B5
Global PD E6912A64457557469E5C691B4D57ABDBBF4C45ADB05492777C574223D0C06F8A
TH1 sphere EE620E6B085A5F9E1C73CFDD1AD04FC0682CEC74DA3DBF8AFE70DD19C038E3A0
TH2 sphere 4D1B627C0343A1C464319704EAFADCA127902A2E4E90CF1C283004359B7ADC24
THXY sphere EABF67C0D1AE309F0281297710A283B8ED5A11D88C8E810837932313F4C53227
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